

DBMS MODEL QUESTION PAPER-1

MODULE - I :-

- 1-a. Compare DBMS and early file systems bringing out the major advantages of the database approach.

Ans. i. Self-describing nature of a database system:- The fundamental characteristic of the DB approach is that the database system contains not only the DB itself but also a complete definition or description of the DB structure & constraints.

- This definition is stored in the DBMS catalog, which contains info such as the structure of each file, the type & storage format of each data item & various constraints on the data. The information stored in the catalog is called meta-data.
- The DBMS catalog will store the definition of all the files in the form of tables.

ii. Insulation b/w program & data, & Data Abstraction:-

- In traditional file processing, the structure of data files is embedded in the application programs, so any change to the structure of a file may require changing all programs that access that file.
- By contrast, DBMS access programs do not require such changes in most cases.
- The structure of data files is stored in the DBMS catalog separated from the access programs. We call this property as program-data independence.

iii. Support of multiple views of the data:

- A DB typically has many users, each of whom may require a agent perspective or view of the DB.

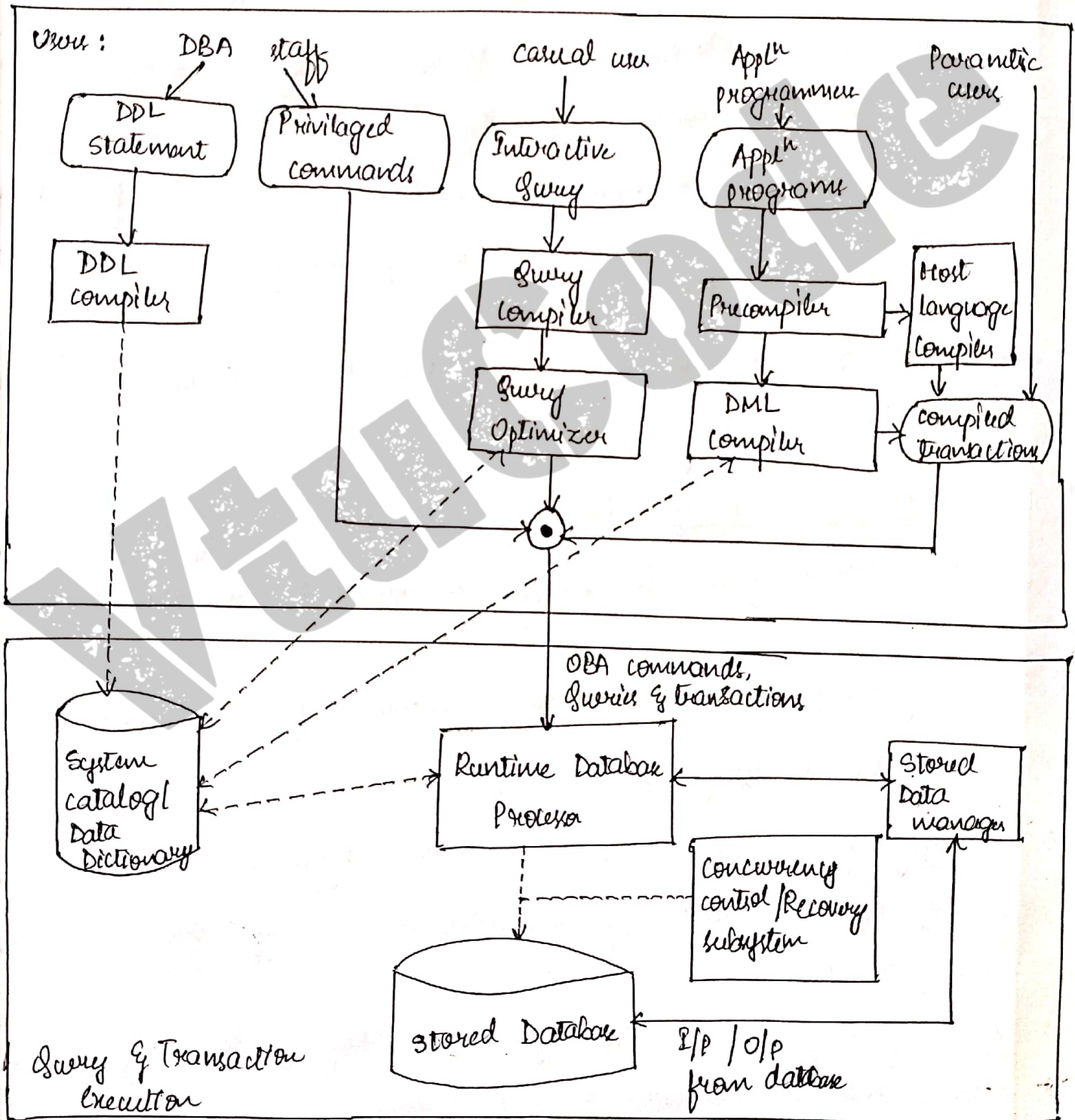
1. Q. - Answer. A view may be a subset of the DB or it may contain virtual data, i.e. derived from the DB files but is not explicitly stored. (2)

- A multuser DBMS whose users have a variety of distinct application must provide facilities for definition of multiple views.

10. Sharing of Data & multuser transaction processing

- A multuser DBMS, as its name implies, must allow multiple users to access the DB at the same time.
- This is essential if data for multiple application is to be integrated and maintained in a single DB.
- When several reservation clerks try to assign a seat on an airline flight, the DBMS should ensure that each seat can be accessed by only one clerk at a time for assignment to a passenger. These types of application are generally called online transaction processing [OLTP] application.
- A fundamental role of multuser DBMS software is to ensure that concurrent transaction operate correctly & efficiently.
- The concept of a transaction has become central to many DB application. A transaction is an executing program or process that includes one or more DB accesses such as loading or updating of DB records.
- Each transaction is supposed to execute a logically correct DB access if executed in its entirety without interference from other transaction.

- 1.6. with a neat block diagram, explain the architecture of a typical DBMS.
- Ans. The figure shows the typical DBMS components. The figure is divided into two halves.
- The top half of the fig. refers to the various users of the DB environment & their interface.
 - The bottom half shows the internals of the DBMS responsible for storage of data & processing of transactions.



1.5. Considering the top-half of the figure.

- It shows interfaces for the DBA staff, casual users who work with interactive interface to formulate queries, appl. programmers who program using some host lang. & parametric users who do data entry by supplying parameters to predefined transactions.
- The DBA staff works on defining the DB & tuning it by making changes to its definition using the DDL statements & other privileged ends.
- The DDL compiler processes schema definition specified in the DDL & stores descriptions of the schemas in the DBMS catalog.
- The catalog includes information such as: the name & size of files, names & data type of data items, storage details of each file & all other information that are needed by the DBMS modules.
- Casual users & persons which need info. from the DB interact using some of interface, i.e.
- Interactive Query interface, these queries are parsed, analyzed for correctness of the operation from the model, the names of data elements & so on.
- A Query compiler, compiles them into an internal form. The internal query is subjected to query optimization.
- A query optimizer; is concerned with rearrangement of possible mode of operation, elimination of redundancy & use of correct & indexes during execution.
- Application programmers write pgm in host languages such as java, COBOL that are submitted to a precompiler.

1.b. - The precompiler extracts DML commands from an application program written in a host programming language. ⑤

- These commands are sent to the DML compiler for compilation into object code for database access.
- The rest of the pgm is sent to the base-lang compiler.
- In the lower half of the figure.
 - The runtime database processor is shown to execute
 - i. The privileged commands ii. The executable query plans iii. The canned transaction with runtime parameters.
 - It works with the system catalog data dictionary & update it with statistics
 - It also works with the stored data manager, which in turn is user basic OS routines for carrying out low-level I/O/output operation b/w disk & main memory

1.c. Explain different types of user friendly interfaces and types of users who typically use each.

Ans. A database management system (DBMS) interface is a user interface which allows for the ability to input queries to a database without using the query language itself.

- i. Menu Based Interface for web clients or browsing:- These interfaces present the user with list of options called menu's, that lead the user through the formulation of a request.
 - By using menus, the user need not to memorise the specific commands & syntax of a query language, rather, the query is composed step-by-step by picking options from a menu.

- ii. Form-Based Interface It displays a form to each user. Users can fill out all the form entries to insert new data, or they can fill out only certain entries in which DBMS will retrieve missing data for the remaining entries.
- iii. Graphical user Interface:- It typically displays a schema to the user in diagrammatic form.

 - The user then can specify a query by manipulating the diagram.
- iv. Natural Language Interface:- These interfaces accept requests written in English or some other language & attempt to understand them.

 - It has its own schema, which is similar to the DB conceptual schema, as well as a dictionary of important words.
- v. Speech I/P & O/P :- Limited use of speech as an I/P query & speech as an answer to a question or result of a request is becoming common place.

 - Application with limited vocabularies such as for telephone dictionary, flight arrival/departures etc are allowing speech for I/P & O/P to enable ordinary folks to access this information.
- vi. Interfaces for Parametric users:- Parametric users such as bank tellers, often have a small set of operation that they must perform repeatedly.
- viii. Interfaces for the DBA:-

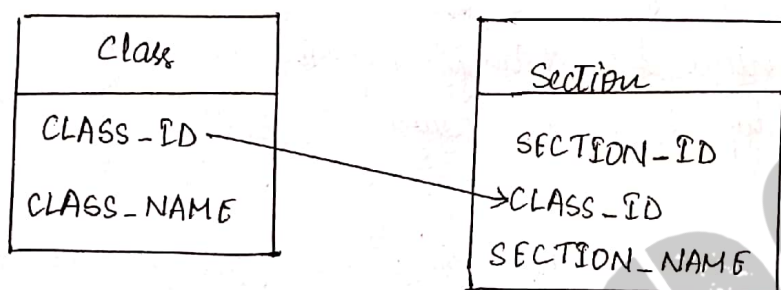
 - Most DB systems contain privileged commands that can be used only by the DBA staff.
 - These include commands for creating accounts, setting system parameters, granting account authorization, changing a schema & reorganizing the storage structure of a DB.

2.a. Define the following with an example:

- i. Weak entity type ii. Participation constraint iii. Cardinality ratio
iv. recursive relationship v. specialization

Ans. i. Weak entity type:- In a relational database, a weak entity is an entity that cannot be uniquely identified by its attributes alone, therefore, it must use a foreign key in conjunction with its attributes to create a primary key. The foreign key is typically a primary key of an entity it is related to.

Ex:-



On considering the CLASS & SECTION entity. The SECTION has SECTION-ID & NAME as its attribute. But SECTION-ID alone cannot be a primary key, since it fails to tell for which course it is related to.

Hence SECTION-ID is a weak-entity type.

ii. Participation constraint:- The participation constraint specifies whether the existence of an entity depends on its being related to another entity via the relationship type.

- This constraint specifies the minimum no. of relationship instances that each entity can participate in & is sometimes called the minimum cardinality constraint.

- The cardinality ratio & participation constraints, taken together as the structural constraints of a relationship type.

2.a

(8)

iii. Cardinality ratio:- The cardinality ratio specifies the maximum number of relationship instances that an entity can participate in.

Ex:- In the WORKS-FOR binary relationship type, DEPARTMENT:EMPLOYEE is of cardinality ratio 1:N, meaning that each department can be related to any number of employees, but an employee can relate to only one department.

iv. Recursive Relationship:- If the same entity type participates more than once in a relationship type in different roles. In such cases, the role name becomes essential for distinguishing the meaning of each participation. Such relationship types are called recursive relationship.

v. Specialization:- Specialization is the process of defining a set of subclasses of an entity type. This entity type is called the superclass of the specialization.

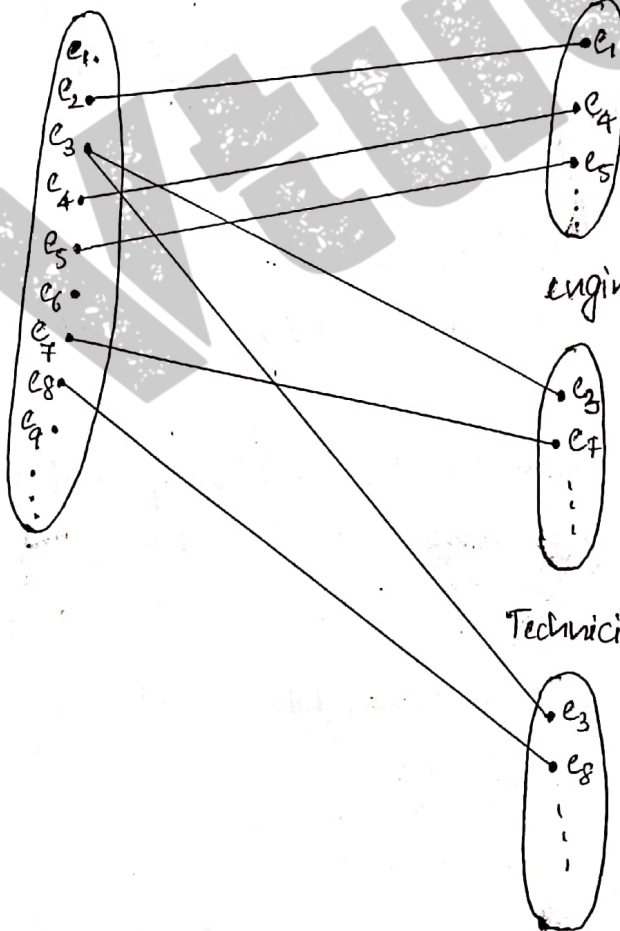
Ex:-

Employee

secretary

engineer

Technician



2.6. Develop an ER diagram for keeping track of information about a company database taking into account atleast five entities.

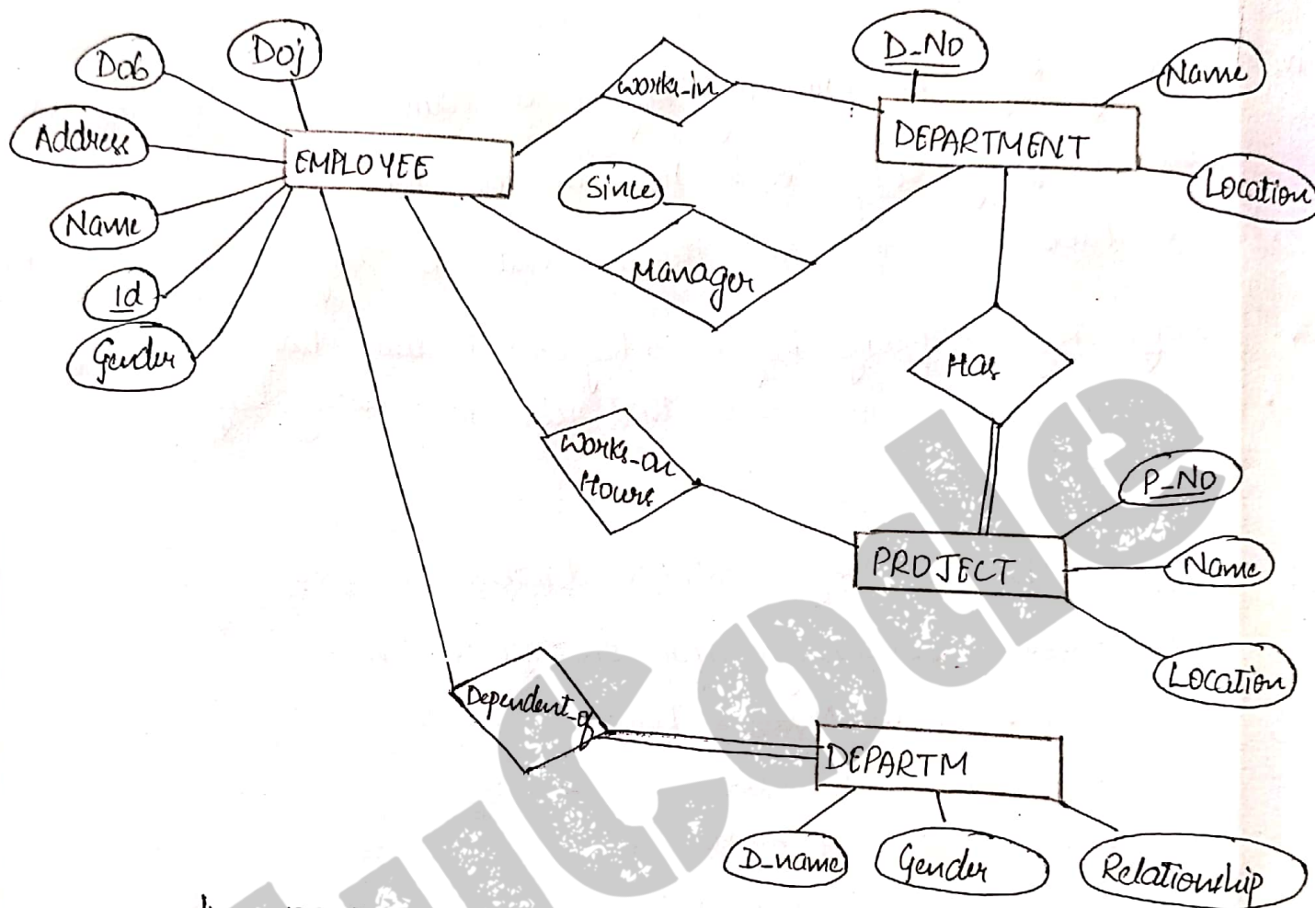


Fig:- ER-Diagram of a Company.

MODULE-2 :-

3.a. Define the following terms,

- i. Key
- ii. Super Key
- iii. Candidate Key
- iv. Primary Key
- v. Foreign Key.

Ans. i. Key:- Key is an attribute or set of attributes which helps in identifying a row (tuple) in a relation (table).

- Keys help uniquely in identifying a row in a table by a combination of one or more columns in that table.

ii. Super Key:- The set of attributes which can uniquely identify a tuple is known as Super Key.

3.a.

iii.

Candidate Key:- Relation Schema may have more than one key. We call such key as, candidate key.

iv.

Primary Key:- The primary key is a column, or set of columns, whose values uniquely identifies each row in the table.

• Each Table in a relational database must be assigned a primary key.

v.

Foreign Key:- A foreign key is a key used to link two tables together.

• It is a field in one table that refers to primary key in another table.

3.b.

Consider the following COMPANY database

EMP (Name, SSN, Salary, Super SSN, Gender, Dno)

DEPT (Dnum, Dname, MgrSSN, Dno)

DEPT-LOC (Dnum, Dlocation)

DEPENDENT (ESSN, Dep-name, Sex)

WORKS_ON (ESSN, Pno, Hours)

PROJECT (Pname, Pnumbn, Plocation, Dnum)

Write the relational algebra queries for the following

i.

Retrieve the name, address, salary of employees who work for the Research department.

ii.

Find the names of employees who work on all projects controlled by department number 4.

iii.

Retrieve the SSN of all employees who either in department no: 4 or directly supervise an employee who work in department number: 4

iv.

Retrieve the names of employees who have no dependents.

v.

Retrieve each department number, the number of employees in the department and their average salary.

36.

(11)

i. Select Fname, Lname, address, salary
 From Employee
 Where Dname = Research;

ii. Select Fname, Lname
 From Employee E, ~~works-on~~ B
 Where Not exists (select *
 From works-on B
 Where (B.Pno In (select Pnumba
 From Project
 Where Dnum=5));
 AND
 NOT EXISTS (select *
 From works-on C
 Where C.ESSN = SSN
 And C.Pno = B.Pno));

iii. Select SSN
 From Employee as E, Employee as S
 Where dno=4, E-superv-ssn = S.SSN;

iv. Select Lname, Fname
 From Employee
 Where not exists (select *
 From department
 Where SSN = essn);

v. Select Dno, count (*), Avg (salary)
 From Employee
 Group by Dno;

3.c. Summarize the steps involved in converting the ER constructs to corresponding relational tables. (12)

Ans. An ER diagram is a pictorial representation of the information that can be captured by a database.

ER diagram consists of

- Entity sets.
- Relationship sets.

Converting each entity/relationship set to a table, which involves deciding the tables

- Attributes
- Candidate Key.

As the ER diagram representation becomes more complex & crowded.

4.a. Explain with example basic constraints that can be specified when a database table is created. (13)

Ans. - NOT NULL :- Ensures that a column cannot have a NULL value.

Ex:-
CREATE TABLE Person (
ID int NOT NULL,
LastName varchar(255) NOT NULL,
FirstName varchar(255) NOT NULL,
Age int);

- UNIQUE :- Ensures that a column cannot have a NULL value.

Ex:-
CREATE TABLE Person (ID int NOT NULL, LastName varchar(255) NOT NULL,
FirstName varchar(255), Age int, UNIQUE (ID));

- PRIMARY KEY :- A combination of NOT NULL and uniquely identifies each row in a table.

Ex:-
Create Table Person (ID int primary key, LastName varchar(255), FirstName
varchar(255), Age int);

- FOREIGN KEY :- It uniquely identifies a row/record in another table.

Ex:-
Create Table Person (ID int primary key, FirstName varchar(20), LastName
varchar(20), sub-id varchar(20), foreign key (sub-id)
references subject (sub-id));

- DEFAULT :- It is used to provide a default value for a column.

Ex:-
Create Table Person (ID int primary key, FirstName varchar(20), LastName
varchar(20), city varchar(20) DEFAULT 'Saudi');

Q.6. Write SQL syntax for the following with example.

- i. SELECT
- ii. ALTER
- iii. UPDATE

Ans:- i. The select operation is used to select a subset of the tuples for a relation that satisfies a selection condition.

- SELECT operation to be a filter that keeps only tuples that satisfy a qualifying condition.

Syntax:- $\sigma_{\langle \text{selection condition} \rangle} (R)$

Ex:- To select the EMPLOYEE tuples whose department is 4, or those salary is greater than \$30,000

$\sigma_{DNO=4} (EMPLOYEE)$
 $\sigma_{salary > 30000} (EMPLOYEE)$

ii. ALTER:- The definition of a base table or of other named schema elements can be changed by using the ALTER command.

- For base tables, the possible alter table actions include adding or dropping column, changing a column definition & adding or dropping table constraints.

Syntax:- $ALTER\ TABLE\ table_name\ ADD\ column_name\ datatype;$

Ex:- $ALTER\ TABLE\ company.Employee\ ADD\ column\ job\ varchar(10);$

4.6.
iii.

(15)

UPDATE :-

- The update operation is used to change the value of one ^{or} more attributes in a tuple of some relation R:-
- It is needed to update the tuples in the table. This command is used to modify attribute values of one or more selected tuples.

Syntax :- 1. update <table-name> set <column-name> = NULL;

2. update <table-name> set <column-name> = 'value' where <column-name> = 'value' condition;

3. To update multiple values,

update <table-name> set <column-name> = 'value' where <column-name> = 'value' and 'value' condition;

Ex:- Update student set marks = NULL;
Update student set marks = 95 where Name = 'xyz';
Update student set marks = 95 where Name = 'xyz' and Name = 'cmu';

4.c. Consider the following relation schema

Works (Pname, Cname, Salary)

Lives (Pname, Street, City)

Located-in (Cname, City)

Manager (Pname, Mgrname)

Write the SQL queries for the following.

- Find the names of all persons who live in the city Bangalore.
- Retrieve the names of all persons of "Infosys" whose salary is between ₹5000.
- Find the names of all persons who live and work in same city.
- List the names of the people who work for "Tech M" along with the cities they live in.
- Find the average salary of "Infosys" persons.

4.c.

Ans. i. Select Fname, Lname
From works
Where City = Bangalore;

ii. Select Fname, Lname
From works
Where Cname = Infosys And salary = 50000;

iii. Select Fname, Lname
From works
Where Lives.city = Located-in.city ;

iv. Select Fname, Lname, city
From works, Lives
Where Cname = Tech M;

v. Select AVG (DISTINCT salary) As "Avg Salary"
From works
Where Cname = Infosys;

5. Q. Explain the following constructs used in SQL with example. (17)

- i. Nested queries ii. Aggregate functions iii. Trigger iv. Views and their updability v. Schema change statements.

Ans. i. Nested query:- Some queries require that existing values in the DB be fetched & then used in a comparison condition, such queries can be conveniently formulated by using nested queries, which are complete select-from-where blocks within the WHERE clause of another query. That other query is called outer query.

Ex:- select Distinct Pnumbr
from Project

where Pnumbr IN

(select Pnumbr

from Project, Department, Employee

where Dnum = Dnumbr AND

mgr_ssn = SSN AND Lname = 'Smith')

ii. Aggregate function:- It is a relational operation, it contains no. of built-in functions: COUNT, SUM, MAX, MIN & AVG.

- The COUNT function returns the no. of tuples or values as specified in a query. The functions SUM, MAX, MIN & AVG are applied to a set or multiset of numeric values.

Ex:- Find the SUM of the salaries of all employees, the maximum salary, the minimum salary & the average salary.

⇒ select sum(salary), max(salary), min(salary), avg(salary)
from Employee;

iii. Trigger: A trigger is a stored procedure in database which automatically invokes whenever a special event in the database occurs.
Ex:- An invoked when a row is inserted into a specified table or when certain table columns are being updated.

iv. Views: - A view in SQL is a single table i.e. derived from other tables. These other tables can be base tables or previously defined views.
- Updating of views is complicated & can be ambiguous. An update on a view defined on a single table without any aggregate function can be mapped to an update on the underlying base table under certain conditions.

Ex:- Consider the WORKS-ON1 view, & suppose that the command to update the PNAME attribute of 'JohnSmith' from 'Product X' to 'Product Y'. This view is updated as:

```
UPDATE WORKS-ON1  
SET Pname = 'Product Y'  
WHERE Lname = 'Smith' AND Fname = 'John'  
AND Pname = 'Product X';
```

v. Schema Change statements: - A schema change is an alteration made to a collection of logical structures in a database. Schema changes are generally made using structured query language and are typically implemented during maintenance windows.

Ex:- dbo, guest, sys, and INFORMATION - SCHEMA.

5.6. What is Dynamic SQL? How it is different from Embedded SQL. (19)

Ans. Dynamic SQL:- It is SQL statements that are constructed at runtime. Dynamic SQL is a programming technique that enables us to build SQL statements dynamically at runtime. We can create more general purpose, flexible applications by using dynamic SQL because the full text of a SQL statement may be unknown at compilation.

- Embedded or static SQL are SQL statements in an application that do not change at runtime and, therefore, can be hard-coded into the application.
- Using static SQL has a benefit which is the optimization of the statements that results in an application with high performance as it offers a good flexibility better than Dynamic SQL.
- Since the access plans for dynamic statements are generated at run-time so they must be prepared in the application, and this is something that will never be looked at in the static SQL.
- Dynamic SQL has one advantage over static statements which can be clearly noticed once the application is edited or upgraded.

5.7. Consider the following COMPANY database (1)

EMP (CName, SSN, Salary, Super SSN, DNO)

DEPT (DNum, Dname, Mgr SSN, Dno)

DEPT_LOC (Dnum, Dlocation)

DEPENDENT (ESSN, Dep-name, Sex)

WORKS-ON (ESSN, PNO, Hours)

PROJECT (Pname, Pnumber, Plocation, Dnum)

- i. Retrieve the name of the employee who works with same department as Ravi.
- ii. Retrieve the number of departments for an employee "Ravi".
- iii. Retrieve the name of the manager working in location "DELHI".

5.c.

Ans.

i. Select Fname, Lname

From Employee E, Department D

Where E.Dno = D.Dno And Fname = Ravi

ii. Select Fname, Lname

From Employee

Where MgrSSN = SSN And DLocation = Delhi And Not exists (select *

From Dependent

Where sex = female And SSN = SSN)

iv. Select *

From employee

Where sex = female And Dno = 20 And Salary > 50000

v. Select *

From department

Where dname = CSE;

6.a. What is SQLJ? How it is different from JDBC.

Ans. SQLJ [Embedding SQL commands in Java]:-

- SQL can be embedded in an object-oriented programming language i.e. the Java language.
- SQLJ is a standard that has been adapted by several vendors for embedding SQL in Java. SQLJ was developed after JDBC, which is used for accessing SQL DB from Java using function calls.
- An SQLJ translator will generally convert SQL statements into Java, which can be executed through the JDBC interface. Hence, it is necessary to install a JDBC driver when using SQLJ.

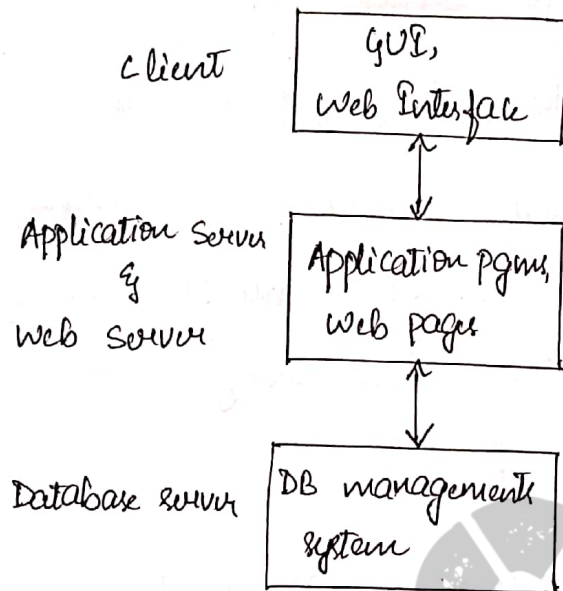
SQLJ different from JDBC:-

- While JDBC provides a complete dynamic SQL interface from Java to relational databases, SQLJ fills a complementary role for static SQL.
- SQLJ source programs are smaller than equivalent JDBC programs because SQLJ provides a shorter syntax.
- SQLJ uses database connections to type-check static SQL code. JDBC being a completely dynamic API, does not do any type checking until run-time.
- SQLJ provides strong typing of query outputs and return parameters and allows type-checking on calls. JDBC passes values to and from SQL without compile-time type checking.
- SQLJ provides simplified rules for calling SQL stored procedures and functions.

(27)

6.b. Draw and explain 3-tier Architecture and Technology relevant to each tier. Write the advantages of 3-tier architecture.

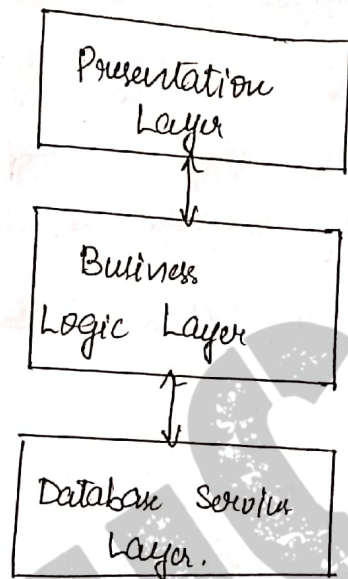
Ans. Many web applications use an architecture called the three-tier architecture, which has an immediate layer b/w the client & the DB server as shown in below figure.:



- The intermediate layer or the middle tier is sometimes called the application server & sometime the webserver, depending on the application.
- This server plays an intermediate role by storing business rules that are used to access data from the DB server.
- It also improves DB security by checking a client's credentials before forwarding a request to the DB server.
- Clients contain GUI interfaces & some additional application specific business rules.
- The intermediate server accepts requests from the client, processes the request and sends DB commands to the DB server, & then acts as a conduit for passing partially processed data from the DB server to the clients, where it may be processed further & filtered to be presented to users in GUI format.

6.b. The below figure shows the other architecture used by DB & other application package vendors. (23)

- The presentation layer displays information to the user, & allows data entry.
- The business logic layer handles intermediate rules & constraints before data is passed up to the user or down to DBMS.
- The bottom layer includes all data mgmt servs. If the bottom layer is split into two layers, then this becomes a four-tier architecture.



6.c. Write a short note on

a. HTML forms b. Java scripts c. CGI d. Application servers e. Servlets.

- Ans.
- a. HTML forms:- An HTML form is a section of a document containing normal content, markup, special elements called controls (check boxes, radio buttons, menus, etc. and labels on those controls.
- Users generally "complete" a form by modifying its controls, before submitting the form to an agent for processing.

6.c.

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6. Java scripts:- JavaScript is a dynamic computer programming language. It is lightweight and most commonly used as a part of web pages, whose implementations allow client-side script to interact with the user and make dynamic pages.
- It is an interpreted programming language with object-oriented capabilities. It is called JavaScript because it was generated by java.
- c. CGI:- In computing, Common Gateway Interface (CGI) is an interface specification for web servers to execute programs like console applications running on a server that generates web pages dynamically.
- Such programs are known as CGI scripts or simply as CGI's. The specifics of how the script is executed by the server are determined by the server.
 - In the common case, a CGI script executes at the time a request is made and generates HTML.
- d. Application servers:- An application server is a server specifically designed to run applications. The "server" includes both hardware and software that provide an environment for programs to run.
- Application servers are used for many purposes.
- e. Servlets:- Servlets are the java programs that run on the java-enabled web server or application server.
- They are used to handle the request obtained from the web server, process the request, produce the response, then send response back to the web server.

MODULE - 4 :-

7.a. Explain insertion, deletion and modification anomalies. Why are they considered bad?

Ans. One goal of schema design is to minimise the storage space that the base relation occupy.

- Grouping attributes into relation schema has a significant effect on storage space.
- Space used by EMP & DEPT as compared with EMP-DEPT is less.
- Another serious problem with using relation EMP-DEPT as base relation is the problem of update anomalies. These can be classified into insertion anomalies, deletion anomalies & modification anomalies.

- Insertion Anomalies:- These can be differentiated into two types,

- To insert a new employee tuple into EMP-DEPT, we must include either the attribute values at the department that the employee works for or null.

Eg:- To insert a new tuple for an employee who works in dept no 2, we must enter attribute values of department 2 correctly so that they are consistent with values for dept 2 in other tuples.

- It is difficult to insert a new department that has no employees. The only way to do this is to place null value in the attribute for employee. This causes problem because EMPID is a primary key.

- Deletion Anomalies:- If we delete from EMP-DEPT an employee tuple that happens to represent the last employee working for a particular department, the information concerning that department is lost from the db.

7.a. This problem will not occur if it is saved separately.

- Modification Anomalies:-

- In EMP-DEPT, if we change the value of one of the attribute of a particular department say from 'CS' to 'comp sci' then we must update the tuple of all employees who work in that department otherwise the db will become inconsistent.

Suppose that we used EMP_PROJ & EMP_LOCS as the base relations instead of EMP-DEPT. This produces a particularly bad schema design, because we cannot recover the information that was original in EMP-DEPT from EMP_PROJ & EMP_LOCS.

7.b. Explain 1NF, 2NF, 3NF with example.

Ans. First Normal Form (1NF):- It states that the domain of an attribute must include only atomic values and that the value of any attribute in a tuple must be a single value from the domain of that attribute.

- So, it disallows multivalued attributes, composite attributes & their combination

Eg:- DEPT

DNAME	DNO	SSN	D-LOCATION
CS	1	1234	{Bang, Mys, Mang}
IS	2	1235	{Ban}

We assume that each department can have a no. of locations. This is not in 1NF, since DLOCATION is not an atomic attribute.

There are three main techniques to achieve 1NF,

1. Remove the attribute DLOCATION that violates 1NF & place it in a separate relation

DEPT-LOCATION

DNO

DLOCATION

1	Bng
1	Mys
1	Man
2	Bng

DEPT

DNAME	DNO	SSN
CS	1	1234
IS	2	1235

2. Expand the key so that there will be a separate tuple in the original DEPT.

DEPT

DNAME	<u>DNO</u>	SSN	<u>DLOCATION</u>
CS	1	1234	Bng
CS	1	1234	Mys
CS	1	1234	Man
IS	2	1235	Bng

3. If maximum no. of value is known for the attribute,
 ex:- It is known that DLOCATION can have atmost three locations.

Second Normal Form:- It is based on the concept of full functional dependency.

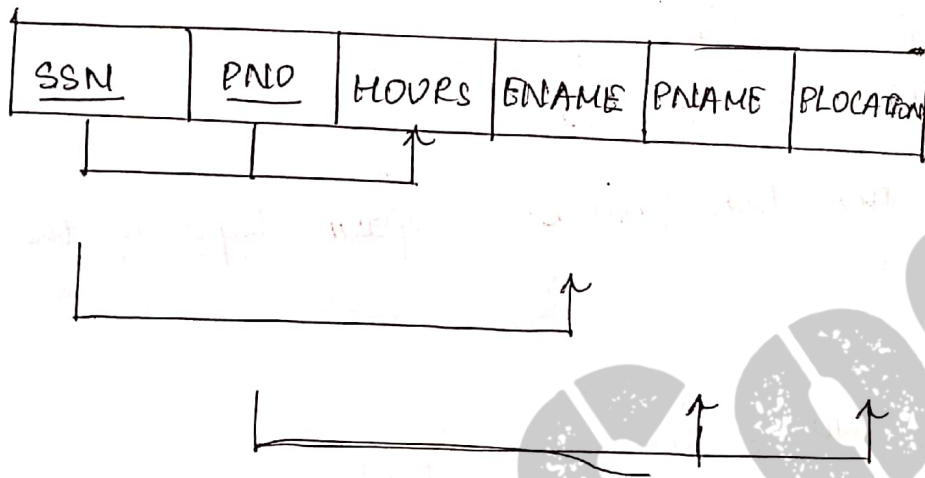
A functional dependency $x \rightarrow y$ is a full functional dependency. If removal of any attribute A from x means,

A functional dependency $x \rightarrow y$ is partial dependency if some attribute AEX can be removed from x & dependency still holds.

- 7.6. $\{SSN, PNO\} \rightarrow HOURS$ is fully dependent
 $\{SSN, PNO\} \rightarrow ENAME$ is partial because
 $SSN \rightarrow ENAME$ holds.

A relation schema R is in 2NF if every nonprime attribute A in R is fully functionally dependent on the primary key of P .

Ex:- EMP_PROJ



$\{SSN, PNO\} \rightarrow HOURS$
 $SSN \rightarrow ENAME$
 $PNO \rightarrow \{PNAME, PLOCATION\}$

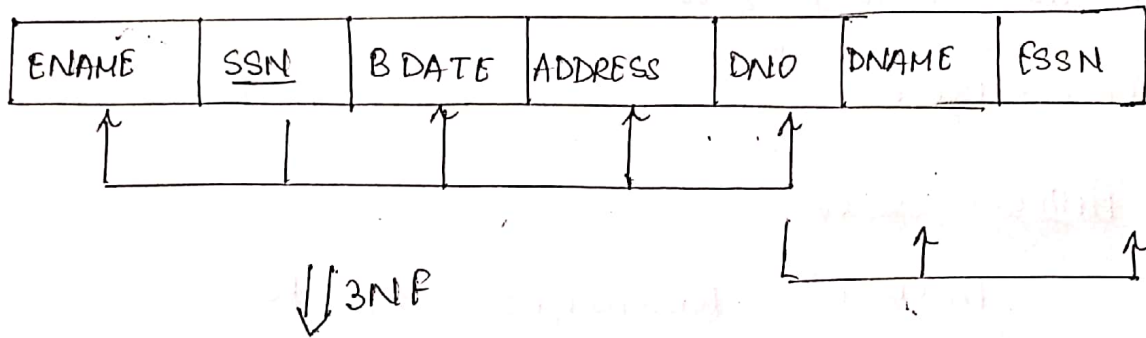
PNO is not full dependent on the key

Third NORMAL FORM (3NF):- It is based on the concept of transitive dependency.

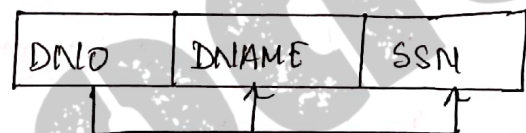
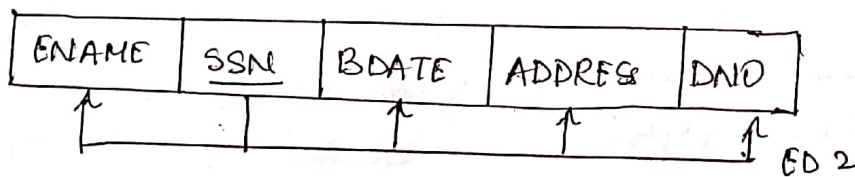
- A functional dependency $X \rightarrow Y$ in a relation schema R is a transitive dependency if there is a set of attributes Z that is neither a candidate key nor a subset of any key of R , & both $X \rightarrow Z$ & $Z \rightarrow Y$ hold.

7.6. A relation schema R is in 3NF if it satisfies 2NF & no non-prime attributes R is transitively dependent on the primary key. (29)

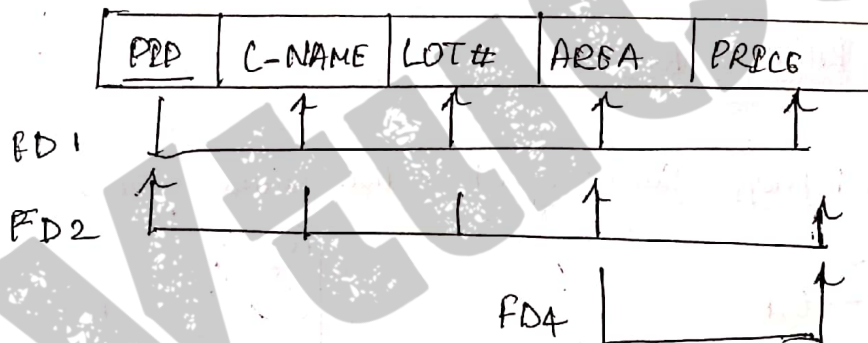
EMP-DEPT



ED1

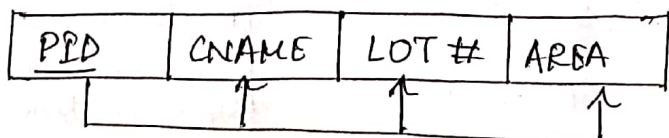


2NF :-

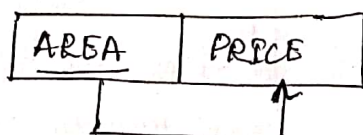


3NF :-

LOTS 1A



LOTS 1B



7.c. Consider the relation schema $R(A, B, C, D, E, F)$ and the functional dependencies $A \rightarrow B, C \rightarrow DF, AC \rightarrow E, D \rightarrow F$. What is the primary key of this relation R ? What is its highest normal form? Preserving the dependencies, decompose R into third normal form.

$\Rightarrow R(A, B, C, D, E, F), F = \{A \rightarrow B, C \rightarrow DF, AC \rightarrow E, D \rightarrow F\}$

Attribute closure:

$$ABCDF^+ = \{A, B, C, D, E, F\} \Rightarrow \text{Super key}$$

$$AC^+ = \{A, C, B, D, F, E\} \Rightarrow \text{Super key}$$

Candidate key:

$$AC^+ = A^+ \Rightarrow \{A, B\}$$

$$C^+ \Rightarrow \{C, D, F\}$$

$\therefore$ the Attribute closure of candidate key is not super key

$\therefore AC$ is a candidate key.

So Prime Attribute are A, C

Non-Prime Attribute are B, D, E, F

$\therefore \underline{AC}$ are the Primary key

2NF: Partial Dependency: Proper subset of CK $\rightarrow$ non-prime attribute

$$AC \Rightarrow \begin{matrix} A \rightarrow B \\ C \rightarrow D, F \end{matrix}$$

Here B, D, F are non-prime attribute

$\therefore$ The Given relation is not in 2NF.

Rules:

- ① It is in 1NF
- ② No partial dependency

3NF: $F = \{A \rightarrow B, C \rightarrow DF, AC \rightarrow E, D \rightarrow F\}$

$$A \rightarrow B \Rightarrow \text{LHS } A^+ = \{A, B\} \neq \text{Super key}$$

$$\text{RHS } B^+ = \{B\} \neq \text{Prime attribute}$$

$\therefore$ The Given relation is not in 3NF

Rules:

- ① It is in 2NF
- ② LHS is Super key
- ③ RHS is Prime attribute

BCNF: $F = \{A \rightarrow B, C \rightarrow DF, AC \rightarrow E, D \rightarrow F\}$

$$A \rightarrow B \Rightarrow \text{LHS } A^+ = \{A, B\} \neq \text{Super key}$$

$\therefore$ The Given relation is not in BCNF.

Rules:

- ① It should be in 3NF
- ② LHS must be Super key

8a. Define non-additive join property of a decomposition & write an algorithm for testing of non-additive join property.

Ans. whenever we decompose a relation schema R into $R_1 = (X \cup Y)$ and $R_2 = (R - Y)$ based on an MVD $X \twoheadrightarrow Y$ that holds in R , the decomposition has the nonadditive join property.

Property NSB:-

The relation schema R_1 & R_2 form a nonadditive join decomposition of R with respect to a set F of Functional & Multivalued dependencies if and only if

$$(R_1 \cap R_2) \twoheadrightarrow (R_1 - R_2)$$

or, by Symmetry, if and only if

$$(R_1 \cap R_2) \twoheadrightarrow (R_2 - R_1).$$

Algorithm for testing of nonadditive join property:-

Input : A universal relation R & a set of Functional & multivalued dependencies F .

- (i) set $D := \{R\}$;
- (ii) while there is a relation schema Q in D that is not in 4NF, do
 - {choose a relation schema Q in D that is not in 4NF;
 - Find a nontrivial MVD $X \twoheadrightarrow Y$ in Q that Violates 4NF;
 - replace Q in D by two relation schemas $(Q - Y)$ and $(X \cup Y)$;
 - }

8.6. Write the algorithm to find the minimal cover for a set of FD's
 Consider $R = \{A, B, C, D, E, F\}$ FD's $\{A \rightarrow C, AC \rightarrow D, E \rightarrow AD, E \rightarrow H\}$ Find the irreducible cover for this set of FD's (minimal cover)

Ans.

$F = A \rightarrow C, AC \rightarrow D, E \rightarrow AD, E \rightarrow H$	$G = A \rightarrow CD, E \rightarrow AH$
$A^+ : ACD$	$A^+ : ACD$
$AC^+ : ACD$	$E^+ : AEDCH$
$E^+ : EAHCD$	

$R(A, C, D, E, H)$, FD: $A \rightarrow C, AC \rightarrow D, E \rightarrow AD, E \rightarrow H$

Find canonical cover

Step-1:- Apply decomposition rule on Rhes

FD: $\{A \rightarrow C, AC \rightarrow D, E \rightarrow A, E \rightarrow D, E \rightarrow H\}$

Step-2:- Remove extraneous attribute

FD: $\{A \rightarrow C, AC \rightarrow D, E \rightarrow A, E \rightarrow D, E \rightarrow H\}$

Step-3:- Redundant FD

FD: $\{A \rightarrow C, AC \rightarrow D, E \rightarrow A, E \rightarrow H\}$

FD: $\{A \rightarrow C, AC \rightarrow D, E \rightarrow A, E \rightarrow H\}$

FD' : $\{A \rightarrow C, AC \rightarrow D, E \rightarrow A, E \rightarrow H\}$

$R = \{A, B, C, D, E, F, G, H, I, J\}$

$F = \{A, B\} \rightarrow \{C\}, \{A\} \rightarrow \{D, E\}, \{B\} \rightarrow \{F\}, \{F\} \rightarrow \{G, H\}, \{D\} \rightarrow \{E, J\}$

Attribute Closure:

$AB \rightarrow \{A, B, C, D, E, F, G, H, I, J\} \Rightarrow SK$

$AB^+ = \{A, B, C, D, E, F, G, H, I, J\} \Rightarrow SK$

8.b. Candidate Key:

$$AB^+ = A^+ : ADEIJ$$

$$B^+ : BFGH$$

AB^- is a candidate key

prime attributes : A, B

Non-prime attributes : C, D, E, F, G, H, I, J

2NF:-

PD: Proper subset of CK

$$AB = A^+ \rightarrow D, E$$

$$B^+ \rightarrow F$$

Here D, E, F are non prime attribute

$\therefore$ The given relation is not in 2NF

3NF:-

$$A, B \rightarrow C$$

$$\text{LHS: } AB \Rightarrow AB^+ = \{AB C D E F G H I J\} = SK$$

[LHS is SK]

RHS: $C \rightarrow$ is not a prime attribute

[RHS is prime attribute]

Here LHS $[AB^+]$ is a SK

$\therefore$ The given relation is not in 3NF

8.c. Given below are two sets of FD's for a relation $R(A, B, C, D, E)$: Are they equivalent? $F = \{A \rightarrow C, AC \rightarrow D, E \rightarrow AD, E \rightarrow H\}$ and $G = \{A \rightarrow CD, E \rightarrow AH\}$

Ans-

$$F = A \rightarrow C \quad AC \rightarrow D, E \rightarrow AD, E \rightarrow H$$

$$G = A \rightarrow CD \quad E \rightarrow AH$$

$$A^+ : ACD$$

$$A^+ : ACD$$

$$AC^+ : ACD$$

$$E^+ = EADCH$$

$$E^+ : EAHCD$$

$$F \subseteq G$$

They are equivalent

MODULE-5:-

Q.2. Briefly discuss on the two phase locking protocol used in concurrency control. How does it guarantee serializability.

Ans. Types of locks and system lock tables:-

- Several types of locks are used in concurrency control. To introduce locking concepts gradually, first we discuss binary lock, which are simple, but are also too restrictive for database concurrency control purposes, and so not using in practice. Then we discuss shared/exclusive locks - also known as read/write locks - which provide more general locking capabilities and are used in practical database locking schemes.
- Binary Locks:- A binary lock can have two states or values: locked and unlocked (or 1 and 0, for simplicity). A distinct lock associated with each database item x . If the value of the lock on x is 1, item x cannot be accessed by a database operation that requests the item. If the value of the lock on x is 0, item x cannot be accessed by a database operation that requests the item. If the value of the lock on x is 0, the item can be accessed when requested, and the lock value is changed to 1.
- Shared/Exclusive Locks:- The preceding binary locking scheme is too restrictive for database items because at most, one transaction can hold a lock on a given item. We should allow several transactions to access the same item x if they all access x for reading purposes only. This is because read operations on the same item by different transactions are not conflicting. However, if a transaction is to write an item x , it must have ~~an~~ exclusive access to x .

9.a. Guaranteeing Serializability by Two-phase locking

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A transaction is said to follow the two-phase locking protocol if all locking operations precede the first unlock operation in the transaction. Such a transaction can be divided into two phases.

Growing (first) phase:- An expanding or growing (first) phase, during which new locks on items can be acquired but none can be released.

Shrinking (second) phase:- A shrinking (second) phase, during which existing locks can be released but no new locks can be acquired.

If lock conversion is allowed, then upgrading of locks must be done during the expanding phase, and downgrading of locks must be done in the shrinking phase. Hence, a read-lock(x) operation that downgrades an already held write lock on x can appear only in the shrinking phase.

9.c. Explain Basic timestamping algorithm.

Ans. The idea for this schema is to order the transactions based on their timestamps. A schedule in which the transactions participate is then serializable, and the only equivalent serial schedule permitted has the transactions in order of their timestamp values. This is called timestamp ordering (TO).

- Whenever some transaction T tries to issue a $\text{read-item}(x)$ or a $\text{write-item}(x)$ operation, the basic TO algorithm compares the timestamp of T with $\text{read-TS}(x)$ and $\text{write-TS}(x)$ to ensure that the timestamp order of transaction execution is not violated. If this order is violated then transaction T is aborted and resubmitted to the system as a new transaction with a new timestamp. If T is aborted and rolled back, any transaction T_1 that may have used a value written by T must also be rolled back. Similarly, any transaction T_2 that may have used a value written by T_1 must also be rolled back, and so on. This effect known as cascading rollback.

1. Whenever a transaction T issues a $\text{write-item}(x)$ operation, the following is checked:
 - a. If $\text{read-TS}(x) > \text{TS}(T)$ or if $\text{write-TS}(x) > \text{TS}(T)$, then abort and roll back T and reject the operation. This should be done because some younger transaction with a timestamp greater than $\text{TS}(T)$ - and hence after T in the timestamp ordering - has already read or written the value of item x before T had a chance to write x , thus violating the timestamp ordering.
 - b. If the condition in part (a) does not occur, then execute the write-item operation of T and set $\text{write-TS}(x)$ to $\text{TS}(T)$.

10.a. Explain multi-version concurrency control protocols.

- Ans. Other protocols for concurrency control keep the old values of a data item when the item is updated.
- These are known as multiversion concurrency control, because several versions of an item are maintained. When a transaction requires access to an item, an appropriate version is chosen to maintain the serializability of the currently executing schedule, if possible. The idea is that some read operations that would be rejected in other techniques can still be accepted by reading an older version of the item to maintain serializability.
 - When a transaction writes an item, it writes a new version and the old version(s) of the item are retained. Some multiversion concurrency control algorithms use the concept of view serializability rather than conflict serializability. An obvious drawback of multiversion techniques is that more storage is needed to maintain multiple versions of the database items.

Multiversion Technique Based on Timestamp Ordering:-

In this method, several versions $x_1, x_2, \dots, x_k$ of each data item x are maintained. For each version, the value of version x_i and the following two timestamps are kept:

1. Read-TS(x_i). The read timestamp (x_i) is the largest of all the timestamps of transactions that have successfully read version x_i .
2. Write-TS(x_i). The write timestamp (x_i) is the largest of all the timestamps. This writes the value of version x_i .

10.6. Write short notes on the following

- i. Transaction rollback and cascading rollback.
- ii. Transaction support in SQL
- iii. Shadow paging
- iv. NO-ONDO/REDO Recovery Based on Deferred Update
- v. Recovery Techniques Based on Immediate Update

1. Transaction rollback and cascading rollback: - If transaction fails for whatever reason after updating the database, but before the transaction commits, it may be necessary to rollback the transaction. If any data item values have been changed by the transaction and written to the database they must be restored to their previous values (BFSMs). If a transaction T is rolled back, any transaction S that has, in the interim, read the value of some data item X written by T must also be rolled back. Similarly, once S is rolled back, any transaction R that has read the value of some data item Y written by S must also be rolled back; and so on. This phenomenon is called cascading rollback.

ii. Transaction support in SQL: - The basic definition of an SQL transaction is similar to our already defined concept of transaction. It is a logical unit of work and is guaranteed to be atomic. A single SQL statement is always considered to be atomic - either it completes execution without an error or it fails & leaves the database unchanged.

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ii. Shadow paging:- This recovery scheme does not require the use of a log in a single-user environment. In a multiuser environment, a log may be needed for the concurrency control method. Shadow paging considers the database to be made up of a number of fixed size disk pages - say, n - for recovery purposes.

iv. NO UNDO/REDO

During transaction execution, the updates are recorded only in the log and in the cache buffers. After the transaction reaches its commit point and the log is force written to disk, the updates are recorded in the database. If the transaction fails before reaching the commit point, there is no need to undo any operation because the transaction has not affected the database on disk in any way. Therefore only REDO type log entries are needed in the log, which include the new value of the item written by a write operation.

v. Recovery Technique based on immediate update:- Provisions must be made for undoing the effect of update operations that have been applied to the database by a failed transaction. This is accomplished by rolling back the transaction and undoing the effect of the transaction write-item operations. Therefore the UNDO-type log entries, which include the old value of the item, must be stored in the log.